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Generalized Multiscale Finite Element Method for Non-Newtonian Fluid Flow in Perforated Domain

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Abstract. In this work, we consider a non-Newtonian fluid flow in perforated domains. Fluid flow in perforated domains have a multiscale nature and solution techniques for such problems require high resolution. In particular, the discretization needs to honor the irregular boundaries of perforations. This gives rise to a fine-scale problems with many degrees of freedom which can be very expensive to solve. In this work, we develop a multiscale approach that attempt to solve such problems on a coarse grid by constructing multiscale basis functions. We follow Generalized Multiscale Finite Element Method (GMsFEM) [1, 2] and develop a multiscale procedure where we identify multiscale basis functions in each coarse block using snapshot space and local spectral problems [3, 4]. We show that with a few basis functions in each coarse block, one can accurately approximate the solution, where each coarse block can contain many small inclusions.

PROBLEM FORMULATION

In this paper, we present a multiscale methods for flow problems in perforated domain. Solving these problems entails a very fine mesh and this results to a large degrees of freedom. For this reason, some types of model reduction techniques are needed. When the perforated domains have some scale separation, the homogenization methods are used to compute macroscopic parameters and construct macroscopic equations. When the perforated media do not have scale separation, more general multiscale approaches are used. Our approach follows a Generalized Multiscale Finite Element Method (GMsFEM) in constructing multiscale basis functions [1, 3, 4]. These multiscale basis functions can be re-used for any input parameter to solve the problem on a coarse grid.

Newtonian fluids characterized by a viscosity that can depends on temperature and does not change with the strain. Only a small group of fluids exhibit such constant viscosity. The large class of fluids whose viscosity changes with the strain rate (the relative flow velocity) are called non-Newtonian fluids [5, 6].

Fluid flow of an incompressible viscous fluid is modeled by the momentum equation and and incompressibility condition

$$\rho\left(\frac{\partial u}{\partial t} + u\nabla u\right) = \rho f + \operatorname{div}\sigma, \quad x \in \Omega^\epsilon, \tag{1}$$

$$\operatorname{div} u = 0, \quad x \in \Omega^\epsilon, \tag{2}$$

where stress and strain tensors σ and ε are given by

$$\sigma = -p\mathcal{I} + 2\mu(\dot{\gamma})\varepsilon, \quad \varepsilon = \frac{1}{2}(\nabla u + \nabla u^T),$$

and for non-Newtonian fluids we have

$$\mu = \mu(\dot{\gamma}, t) \quad \dot{\gamma} = |\varepsilon| = \sqrt{\varepsilon(u) : \varepsilon(u)}.$$

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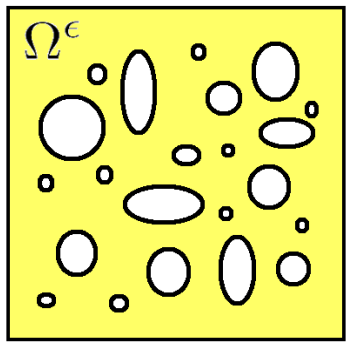

FIGURE 1. Perforated domain, Ω^ϵ

Here $\Omega^\epsilon = \Omega \backslash \mathcal{B}^\epsilon$ is a perforated domain and $\mathcal{B}^\epsilon$ denotes the perforations, see Figure 1 for an illustration.

For slow fluid flow, we consider following Stokes problem

$$\begin{aligned} -\nabla \cdot (\mu \nabla u) + \operatorname{grad} p &= f, \quad x \in \Omega^\epsilon, \\ \operatorname{div} u &= 0, \quad x \in \Omega^\epsilon, \end{aligned} \tag{3}$$

with Dirichlet boundary condition on global boundary $u = g, x \in \partial\Omega \cap \partial\Omega^\epsilon$ and on perforations $u = 0, x \in \partial\Omega^\epsilon \cap \partial\mathcal{B}^\epsilon$.

Here for non-Newtonian fluid we use Carreau model for viscosity

$$\mu = \mu_\infty + \frac{\mu_0 - \mu_\infty}{(1 + (\dot{\gamma}\lambda)^2)^{\frac{1-n}{2}}},$$

where μ_0 and μ_∞ are viscosities at low and high shear rates, λ is characteristic relaxation time, n is power law parameter and for $n = 1$ we have Newtonian fluid flow.

FINE-SCALE APPROXIMATION

For solution of the nonlinear problem (3), we use the Picard iteration. We use a known, previous solution in the coefficient μ and in every Picard iteration solve following linear problem

$$\begin{aligned} -\nabla \cdot (\mu(u^k) \nabla u^{k+1}) + \operatorname{grad} p^{k+1} &= f \quad x \in \Omega^\epsilon, \\ \operatorname{div} u^{k+1} &= 0 \quad x \in \Omega^\epsilon, \end{aligned} \tag{4}$$

where k is the number of nonlinear iteration.

For fine-scale approximation we use finite element method and write variational formulation of the problem (4) as follows: find $u^{k+1} = u \in V$ and $p \in Q$ such that

$$\begin{aligned} a(u, v) + b(v, p) &= (f, v) \quad \forall v \in V \\ b(u, q) &= 0 \quad \forall q \in Q, \end{aligned} \tag{5}$$

where

$$a(u, v) = (\mu \nabla u, \nabla v), \quad b(v, p) = -(\operatorname{div} v, p).$$

Here

$$V_0 = \{v \in [H^1(\Omega)]^d : v = 0, \quad x \in \partial\Omega\}, \quad V = \{v \in [H^1(\Omega)]^d : v = g, \quad x \in \partial\Omega\},$$

and

$$Q = \{q \in L_2(\Omega) : \int_\Omega q \, dx = 0\}.$$

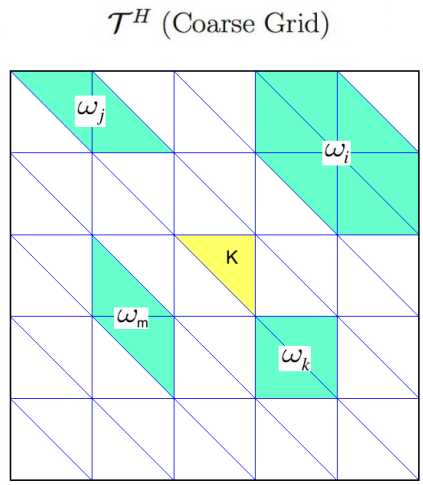

FIGURE 2. Illustration of a coarse grid ($\mathcal{T}^H$), coarse neighborhood (ω_i) and coarse grid blocks (K)

For the approximation on the fine grid, we assume that the computational domain is triangulated so that all perforations are resolved, and we use the standard P^2 space for velocity and the piecewise constant space for pressure. We denote these spaces by $V_h \subset V$ and $Q_h \subset Q$ respectively. The fine-grid solution $(u_h, p_h) \in V_h \times Q_h$ is obtained by solving

$$\begin{aligned} a(u_h, v_h) + b(v_h, p_h) &= (f, v_h) \quad \forall v_h \in V_h, \\ b(u_h, q_h) &= 0 \quad \forall q_h \in Q_h. \end{aligned} \tag{6}$$

We remark that we use the same notation (u_h, p_h) for the approximate solution and for the corresponding coefficients in the expansion of the solution in terms of the basis functions. It is well known that the fine scale solver described above is stable and convergent.

COARSE-SCALE APPROXIMATION

Let $\mathcal{T}^H$ be a standard partition (coarse grid) of the computational domain Ω into finite elements, where H is the coarse mesh size. The fine-grid, $\mathcal{T}^h$ with mesh size h, is a conforming refinement of $\mathcal{T}^H$. Typically $0 < h \ll H$ and the triangulation $\mathcal{T}^h$ is fine enough to fully resolve the perforations. We define the node-based and edge-based coarse neighborhoods ω_i. See Figure 2 for an illustration of coarse neighborhoods and coarse grid blocks subordinated to the coarse discretization. We emphasize that we use ω_i to denote a coarse neighborhood, and use K to denote a coarse grid block.

Next we give an overview of GMsFEM. In general, the following are the main procedures:

1. Coarse grid generation.
2. Construction of an offline space:
   - Construction of a snapshot space that will be used to compute an offline space.
   - Construction of a small dimensional offline space by performing a dimension reduction in the snapshot space.
3. Solution of a coarse-grid problem for any forcing term and boundary condition.

In this paper, we will construct an offline space for the velocity:

$$V_H := \text{span}\{\psi_{i,x}, \psi_{i,y}\}_{i=1}^{N_u},$$

where $\psi_{i,x}, \psi_{i,y}$ are the multiscale basis functions in coarse neighborhood ω_i and N_u is the number of basis functions. For the pressure, we will use the space of piecewise constant functions Q_H over the coarse triangulation $\mathcal{T}^H$. We define the coarse scale system as follows: find $u_{ms} \in V_H$ and $p_{ms} \in Q_H$ such that

$$\begin{aligned} a(u_{ms}, v) + b(v, p_{ms}) &= (f, v), \quad \forall v \in V_H \\ b(u_{ms}, q) &= 0, \quad \forall q \in Q_H, \end{aligned} \tag{7}$$

where

$$Q_H = \{q \in L^2(\Omega) : \ q|_K \in P^0(K), \ \forall K \in \mathcal{T}_H\}.$$

We remark that each ψ_i is supported in the coarse neighborhood ω_i and $N_p = \dim(Q_H)$ is equal to the number of coarse grid blocks.

Next, we will consider the construction of the offline space for the velocity. We first will consider snapshot space that contains an extensive set of basis functions formed by the solution of local problems with all possible boundary conditions up to the fine grid resolution. Then we will apply a spectral decomposition to select dominant modes in the snapshot space, multiply to partition of unity functions (for performing conforming elements) and solve local problem for making divergent constant multiscale basis functions

Snapshot space

Snapshot space is a space, which contains an extensive set of basis functions that are solutions of local problems with all possible boundary conditions up to fine-grid resolution. We solve following problem on the coarse neighborhood ω_i: Find (u_l^i, p_l^i) such that

$$\begin{aligned} \int_{\omega_i} (\nabla u_l^i, \nabla v)\, dx - \int_{\omega_i} p_l^i \operatorname{div} v\, dx &= 0, \quad v \in V_h^i, \\ \int_{\omega_i} q \operatorname{div} u_l^i\, dx &= \int_{\omega_i} c\, q\, dx, \quad q \in Q_h^i, \end{aligned} \tag{8}$$

with boundary condition

$$u_l^i = (0,0), x \in \partial\mathcal{B}^\epsilon, \quad u_l^i = (\delta_l^i, 0) \quad \text{or} \quad (0, \delta_l^i), x \in \partial\omega_i / \partial\mathcal{B}^\epsilon$$

where function δ_l^i is a piecewise constant function defined on E_i such that it has value 1 on e_l and value 0 on the other fine-grid edges. Notice that $\partial\omega_i/\partial\mathcal{B}^\epsilon = \cup_{l=1}^{J_i} e_l$, where e_l are the fine-grid edges and J_i is the number of these fine grid edges on $\partial\omega_i/\partial\mathcal{B}^\epsilon$.

We remark that the constant c in Eq. (8) is chosen by compatibility condition, $c = \frac{1}{|\omega_i|} \int_{\partial\omega_i/\partial\mathcal{B}^\epsilon} u_l^i \cdot n_i\, ds$. The above local problem is solved numerically on the fine-grid defined in ω_i by the lowest-order Crouzeix-Raviart element.

The collection of the solutions of above local problems generates the snapshot space, $\Psi_l^{i,snap} = u_l^i$ in ω_i:

$$R_{i,snap} = \left[\Psi_1^{i,snap}, \ldots, \Psi_{J_i}^{i,snap}\right], \quad V_{snap} = \{\Psi_l^{i,snap} : 1 \le l \le 2J_i, \ 1 \le i \le (N_e + N_v)\},$$

where N_e is the number of coarse-grid edges (faces) and N_v is the number of coarse-grid vertices.

Spectral problem for space reduction

We perform a space reduction on the snapshot space through the use of a local spectral problem in ω_i. The purpose of this is to determine the dominant modes in the snapshot space and to obtain a small dimension space for the approximation the solution.

We consider following local eigenvalue problem in the snapshot space

$$A^{i,off}\psi_k = \lambda_k^{i,off} S^{i,off}\psi_k, \tag{9}$$

where

$$A^{i,off} = [a^{off}_{mn}] = a_i(\Psi^{i,snap}_m, \Psi^{i,snap}_n) = R_{i,snap} A^i_f R^T_{i,snap}, \quad S^{i,off} = [m^{off}_{mn}] = s_i(\Psi^{i,snap}_m, \Psi^{i,snap}_n) = R_{i,snap} M_f R^T_{i,snap}$$

and

$$a_i(u,v) = \int_{\omega_i} (\nabla u, \nabla v) dx, \quad s_i(u,v) = \int_{\omega_i} u\, v\, dx + \int_{\omega_i} \operatorname{div} u \operatorname{div} v dx.$$

Note that the above spectral problem is solved in the local snapshot space corresponding to the neighborhood domain ω_i. We arrange the eigenvalues in increasing order, and choose the first $M_{i,off}$ eigenvalues and take the corresponding eigenvectors $\Psi^{i,off}_k = R_{i,snap}\psi_k$, for $k = 1, 2, ..., M_{i,off}$, to be the basis functions.

Multiscale space

For construction of conforming offline space, we need to multiply our eigenvectors to $\Psi^{i,off}_k$ to partition of unity functions $\chi \in P^2$ and for performing divergence constant multiscale basis functions we again solve two local problems for two dimensional case in $\forall K^i_j \in \omega_i$:

$$\begin{aligned} -\Delta \Phi^{i,off}_{x,k} + \operatorname{grad} p &= 0, \quad x \in K^i_j, \\ \operatorname{div} \Phi^{i,off}_{x,k} &= \frac{1}{|K^i_j|} \int_{\partial K^i_j} (\chi_i \Psi^{i,off}_{x,k}, 0) \cdot n_i\, ds \quad x \in K^i_j, \end{aligned} \tag{10}$$

with boundary condition $\Phi^{i,off}_{x,k} = (\chi_i \Psi^{i,off}_{x,k}, 0)$, $x \in \partial K^i_j$, and

$$\begin{aligned} -\Delta \Phi^{i,off}_{y,k} + \operatorname{grad} p &= 0, x \in K^i_j, \\ \operatorname{div} \Phi^{i,off}_{y,k} &= \frac{1}{|K^i_j|} \int_{\partial K^i_j} (0, \chi_i \Psi^{i,off}_{y,k}) \cdot n_i\, ds \quad x \in K^i_j, \end{aligned} \tag{11}$$

with boundary condition $\Phi^{i,off}_{y,k} = (0, \chi_i \Psi^{i,off}_{y,k})$, on ∂K^i_j.

After solving (10) and (11), we create the offline space by these functions:

$$V_{off} = \operatorname{span}\{\Phi^{i,off}_{x,k},\ \Phi^{i,off}_{y,k} :\ 1 \le i \le (N_e + N_v),\ 1 \le k \le M_{i,off}\},$$

$$R_{i,off} = \left[\Phi^{i,off}_{x,1}, \Phi^{i,off}_{y,1}, \ldots, \Phi^{i,off}_{x,M_{i,off}}, \Phi^{i,off}_{y,M_{i,off}}\right].$$

Using a single index notation, we can write

$$V_{off} = \operatorname{span}\{\Phi^{off}_i\}_{i=1}^{N_u}, \quad R_u = \left[\Phi^{off}_1, \ldots, \Phi^{off}_{N_u}\right],$$

where $N_u = \sum_{i=1}^{2(N_e+N_v)} M_{i,off}$. This space will be used as the approximation space for the velocity.

NUMERICAL RESULTS

In this section, we present simulation results using the mixed GMsFEM framework and compare our results with a reference solution computed in the fine grid. We consider the Stokes problem with zero velocity on the holes $\partial\Omega^\epsilon \cap \partial\mathcal{B}^\epsilon$, $f = (0,0)$ and $u = (1,0)$ on $\partial\Omega$. For numerical simulations we use Carreau model for viscosity with $\mu_0 = 1.0$, $\mu_\infty = 0$, $\lambda = 0.1$ and different power law parameter $n = 0.2, 0.8, 1.0$ (Newtonian flow), 1.2 and 1.8.

We set $\Omega = [0,1] \times [0,1]$ and use three types of perforated domains, where the perforated regions $\mathcal{B}^\epsilon$ are circular and have different sizes. The computational domain is discretized coarsely using uniform triangulation as shown in the

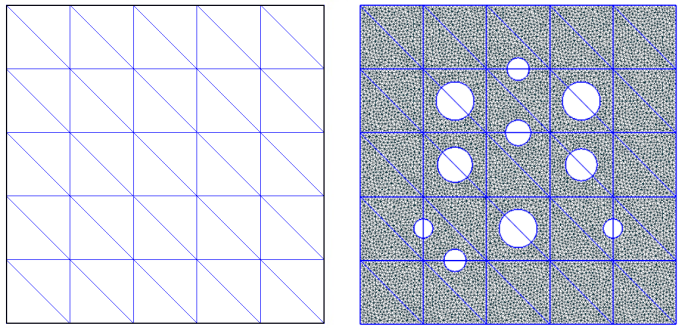

FIGURE 3. Coarse and fine grids for perforated domain

Figure 3, where the coarse mesh size $H = 1/5$. Here DOF_f and DOF_c are sizes of fine-scale and coarse-scale systems. Furthermore, nonuniform triangulation is used inside each coarse triangle element to obtain a finer discretization.

We will consider simulation in perforated domain that illustrated in Figure 3. The coarse grid consists of 50 cells, 85 edges and 36 vertices and the fine grid contains 17912 cells, 27165 edges and 9244 vertices (Figure 3). For Picard iterations we used $\epsilon = 10^{-3}$.

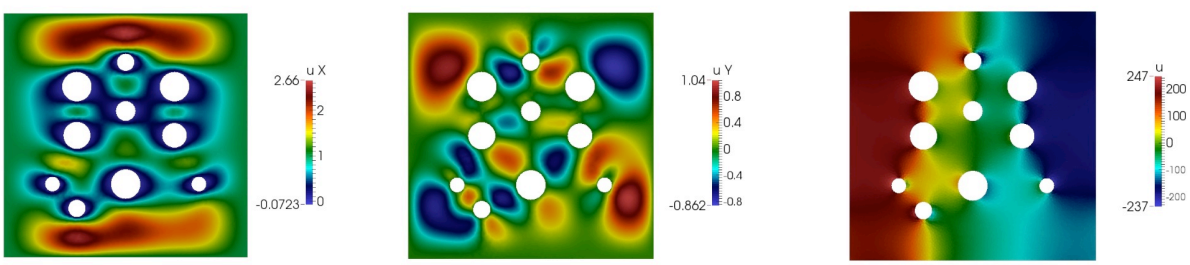

FIGURE 4. Velocity and pressure for $n = 1.8$ for perforated domain. Fine-scale solution, $DOF_f = 72242$

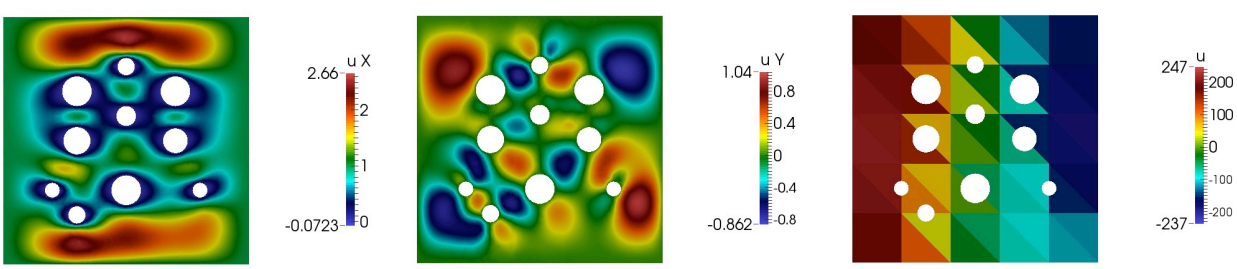

FIGURE 5. Velocity and pressure for $n = 1.8$ for perforated domain. Coarse-scale solution, $DOF_c = 1018$

The errors will be measured in relative L^2 and H^1 norms for velocity and L^2 norm for pressure

$$\|\varepsilon_u\|_{L^2(\Omega^\epsilon)} = \frac{\left(\int\limits_{\Omega^\epsilon}(u_f - u_{ms})^2 dx\right)^{1/2}}{\left(\int\limits_{\Omega} u_f^2 dx\right)^{1/2}}, \quad \|\varepsilon_u\|_{H^1(\Omega^\epsilon)} = \frac{\left(\int\limits_{\Omega^\epsilon}\nabla(u_f - u_{ms})^2 dx\right)^{1/2}}{\left(\int\limits_{\Omega}\nabla(u_f)^2 dx\right)^{1/2}}, \quad \|\varepsilon_p\|_{L^2(\Omega^\epsilon)} = \frac{\left(\int\limits_{\Omega^\epsilon}(p_f - p_{ms})^2 dx\right)^{1/2}}{\left(\int\limits_{\Omega} p_f^2 dx\right)^{1/2}},$$

TABLE 1. Relative errors for velocity and pressure for perforated domain for different $n = 0.2, 0.8, 1.0, 1.2$ and 1.8

M_{off}	DOF_c	$\|\varepsilon_u\|_{L^2(\Omega)}(\%)$	$\|\varepsilon_u\|_{H^1(\Omega)}(\%)$	$\|\varepsilon_p\|_{L^2(\Omega)}(\%)$	$\|\varepsilon_{\bar{p}}\|_{L^2(\Omega)}(\%)$
$n = 0.2$					
2(4)	534	12.864	46.605	31.282	21.762
3(6)	776	3.448	24.906	23.525	4.964
4(8)	1018	2.787	22.578	23.307	3.719
6(12)	1502	2.732	22.233	23.279	3.531
8(16)	1986	2.729	22.205	23.276	3.507
$n = 0.8$					
2(4)	534	11.732	41.811	29.114	18.152
3(6)	776	1.473	10.267	23.179	1.239
4(8)	1018	0.547	5.011	23.150	0.381
6(12)	1502	0.424	4.151	23.148	0.236
8(16)	1986	0.399	3.620	23.148	0.229
$n = 1.0$					
2(4)	534	11.698	41.976	29.547	18.748
3(6)	776	1.424	9.871	23.277	1.162
4(8)	1018	0.398	3.837	23.251	0.270
6(12)	1502	0.148	1.532	23.250	0.114
8(16)	1986	0.135	1.294	23.249	0.104
$n = 1.2$					
2(4)	534	11.725	42.513	30.279	19.807
3(6)	776	1.518	10.726	23.401	1.344
4(8)	1018	0.529	5.156	23.368	0.378
6(12)	1502	0.351	3.555	23.366	0.200
8(16)	1986	0.346	3.449	23.366	0.190
$n = 1.8$					
2(4)	534	2.033	43.902	34.215	25.279
3(6)	776	1.875	13.036	23.974	2.687
4(8)	1018	0.884	8.201	23.863	1.257
6(12)	1502	0.714	6.907	23.849	0.948
8(16)	1986	0.709	6.844	23.849	0.937

where (u_f, p_f) and (u_{ms}, p_{ms}) are fine-scale and coarse-scale using GMsFEM solutions, respectively for velocity and pressure. In addition to the relative L^2 norm for pressure, $\|p_f - p_{ms}\|_{L^2(\Omega)}$, we also compare the numerical pressure with the cell average of the fine scale pressure, that is, $\bar{p}_f = \frac{1}{|K|}\int_K p_f$ for all $K \in \mathcal{T}_H$ ($\|\varepsilon_{\bar{p}}\|_{L^2(\Omega)}$).

In Table 1 we present errors for different power law of parameter $n = 0.2, 0.8, 1.0, 1.2$ and 1.8. Fine-scale solution and coarse scale solutions are presented in Figure 4 and 5, respectively. The fine-scale reference solution is obtained in a space of dimension 72242 for perforated domain and coarse-scale solution is performed using 4 multiscale basis functions per local domain with $DOF_c = 1018$. Compared the fine-scale solution on the left with the coarse-scale

solution on the right of the figures, we observe that the GMsFEM can approximate the fine-scale solution accurately.

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